

Call Analysis - Roadmap and Milestones (v2)

What changed from v1

- **R0 is an internal foundation milestone** (data model, taxonomy, set configuration, de-identification). It delivers nothing user-facing; the first business value lands in R1.
- **Findings attach to the interaction, not the case.** One analysis record per interaction, with one record per question asked on the call. Closes open item O2.
- **Taxonomy is a single tree of arbitrary depth:** caller type (member / provider / prospect) at the top, then category, subcategory, then leaf-specific drill-downs. Member and provider have separate subtrees. Replaces the flat category / subcategory model.
- **Static schema is a hard constraint.** What is extracted per leaf changes in prompts and configuration, never in DDL. Drill-down results are stored in a flexible structure (see Design decision).
- **Enterprise data is not replicated.** Group, subgroup, program type, member data stay external. Analytics tables feed Snowflake, and all reporting, trending and later chat-with-data run from there.
- **New capabilities:** PHI de-identification of transcripts, basic taxonomy editor in R1, targeted (branch-scoped) reprocessing, chatbot outcome tracking split from channel extension, chat-with-data aligned with the Snowflake semantic-layer initiative.
- **Call-level metrics** added: sentiment, questions per call, patients per provider call, captured on the interaction analysis record.

Release overview

Release	Goal	Deliverables	Done when
R0 - Foundation (internal)	Settle the design and the taxonomy so R1 builds on a stable base. No user-facing deliverable.	Analytics data model with static schema and drill-down storage decision; taxonomy v1 (member and provider trees) validated with the business; CareSource Ohio sets and opt-out list configured; PHI de-identification in place.	Data model signed off with reporting engineering; taxonomy v1 approved by the business; de-identified sample set available.
R1 - Capture and static reports	Every interaction categorized against the agreed taxonomy as it arrives; the reports already requested come out of the system.	Categorization pipeline on Call Summarization with call-level metrics; basic taxonomy editor; Snowflake feed; CareSource Ohio PCIR and PCDR; per-group category report.	Manual PCIR/PCDR process retired; per-group report sent to a customer; 100% of new interactions categorized.
R2 - Actionable insight	Move from 'what they called about' to 'what exactly they asked'; let people explore the data.	Leaf-specific drill-down extraction (eligibility/benefits first, then provider search, then toolkit); targeted reprocessing; interactive reporting UI and dashboards; Power BI / Tableau on Snowflake with derived trend measures.	Business confirms drill-downs are actionable for the first three subcategories; a hotspot question ('what about benefits?') is answered from a dashboard.
R3 - Curate and discover	Business owns the taxonomy; the system surfaces what we did not think to look for; chatbot sessions included.	Business admin UI for sets, versions and applicability; clustering-based category discovery; chatbot session categorization with outcome tracking.	A new taxonomy version is published by the business without a feature-team change; first discovered category adopted.
R4 - Explore	Ask any question of the data.	Chat-with-data over Snowflake (semantic layer); ad-hoc mining and keyword search; routing / misroute and call-flow effectiveness; predictive use for conversions; e-mail channel.	A stakeholder answers an unplanned question without a request to IT.

R0 through R2 are well understood. R3 and R4 are directional and will be re-planned once R2 data is in hand; the roadmap is iterative by design, not a big-bang commitment.

Capability mapping to releases

ID	Capability	Release	Note / how it changed
F1	Continuous categorization of every interaction	R1	Pipeline step after Call Summarization; one analysis record per interaction, one record per question.
F2	Seed taxonomy	R0	Now a tree: caller type > category > subcategory > leaf drill-downs; separate member and provider subtrees.
F3	Category sets by payer / plan / group / subgroup	R0	Standard tree applies to all; client sets (CareSource Ohio) run additionally where configured. A call can be categorized under more than one set.
F4	Versioned category sets	R0	Every finding stores the set version it was classified under.
F5	Per-subcategory drill-down extraction	R2	Leaf-specific prompts; eligibility/benefits first. Storage design lands in R0, extraction itself in R2.
F6	Association with interaction metadata	R1 (narrowed)	Caller type copied onto the analysis record for convenience. Group, subgroup, program type are joined in Snowflake, not stored in analytics tables.
F7	Canned per-group report	R1	Category / subcategory percentages per group, from Snowflake.
F8	CareSource Ohio PCIR / PCDR	R1	Confirmed in scope. Separate configured set; provider-only; inquiry vs dispute as category, their list as subcategories.
F9	Data externalized and joinable	R0 / R1	Tables designed in R0; Snowflake feed built in R1; reporting tools read Snowflake.
F10	Evolving extraction and reprocessing	R2	Reprocessing is branch-scoped: rerun only interactions under the changed node.
F11	Category discovery (clustering)	R3	Candidate categories surfaced for curation.
F12	Business-administered taxonomy	R3	Full admin UI. A basic internal editor ships in R1 (F25).
F13	Power BI / Tableau support	R2	On Snowflake; trend measures derived in ETL.
F14	Ad-hoc data mining / keyword search	R4	
F15	Conversational analytics (chat with data)	R4	Aligned with the Snowflake semantic-layer initiative; this project is a concrete use case for it.
F16	Dashboards	R2	Interactive drill-down dashboards over Snowflake.
F17	Misroute and cross-plan analysis	R4	Needs original-queue export from Genesys.
F18	Call-flow / routing change effectiveness	R4	Definition still open (O4).
F19	Predictive use for conversions	R4	
F20	Additional channels	R3 chatbot / R4 e-mail	Chatbot sessions in R3 together with F29; e-mail in R4.
F21	Client opt-out exclusion	R0	Plans that do not approve AI processing are skipped from day one.
F22	Case vs interaction anchoring	R0 (resolved)	Interaction. Case mapping is not built; can be added later if a consumer needs it.
F23	Genesys roadmap check	Not a release item	Proceed regardless; an informational meeting may be scheduled.
F24	New - PHI de-identification of transcripts	R0	Redaction step (Azure service candidate) producing de-identified transcripts for analysis, prompt design and AI-assisted drill-down design.
F25	New - Basic taxonomy editor	R1	Internal tree editor (expand / add / retire nodes, publish version). Business-facing version is F12 in R3.
F26	New - Multi-stage prompting and transcript chunking	R1 / R2	Caller type from interaction; category then subcategory prompts (R1); send only the relevant transcript section to leaf prompts (R2) to control cost.
F27	New - Snowflake analytics feed	R1	ETL of analytics tables including flattening of drill-down structures; derived measures and trends in R2.
F28	New - Primary category per interaction	R1	Alongside all questions found; requested for headline reporting.
F29	New - Chatbot outcome tracking	R3	Did the bot fulfil the request; map to tool calls where possible.
F30	New - Call-level metrics	R1	Sentiment, number of questions per call, number of patients discussed on a provider call; written on the interaction analysis record by the pipeline, reported from R2.

What has to be built, by release

R0 - Foundation (internal)

ID	Build item	Scope	Capabilities	Depends on	Estimate
B1	Analytics data model	Interaction analysis, question/finding, taxonomy node (parent-child), category set, set version, set configuration, exclusion list. Static schema; drill-down payload in a flexible structure (see Design decision). Reviewed with reporting engineering; walked through with sample data.	F1, F4, F9, F22	Design decision below; reporting engineering (Wendy); DBA	
B2	Taxonomy v1 and validation	First-pass member and provider trees. Validate by running sample questions through Copilot ('which node?'), then a business session on eligibility/benefits, then provider search. Publish as version 1.	F2, F3	Business reviewers (Linda, Ashley, Carmen); de-identified sample calls (B5)	
B3	Set configuration and opt-out	Load CareSource Ohio PCIR and PCDR sets with applicability (payer/plan); maintain the list of plans excluded from AI processing.	F3, F8, F21	CareSource category lists; opt-out confirmation	
B5	PHI de-identification	Redaction service over transcripts (SSN, names, IDs) producing a de-identified copy for taxonomy work, prompt tuning and AI-assisted analysis.	F24	Azure redaction service evaluation; security sign-off	

R1 - Capture and static reports

ID	Build item	Scope	Capabilities	Depends on	Estimate
B4	Categorization pipeline	Hook after Call Summarization. Caller type from interaction; prompts for category then subcategory; applies the standard tree plus any configured client set; stores set version, primary category and call-level metrics (sentiment, question count, patients discussed); skips excluded plans. Handles ~1,500+ interactions/day.	F1, F3, F4, F21, F26, F28, F30	B1, B2, B3; LLM capacity; summarization team	
B6	Basic taxonomy editor	Internal UI: expandable tree, add/edit/retire nodes (edits create new rows), publish a version. No business-facing workflow yet.	F25, F4	B1; authentication	
B7	Snowflake feed	ETL of analytics tables to Snowflake, flattening the drill-down structure; join keys to interaction, group, subgroup available on the warehouse side.	F9, F27, F6	Data warehouse team; Snowflake environment	
B8	Static reports	CareSource Ohio PCIR and PCDR (monthly); per-group category/subcategory report; produced from Snowflake.	F7, F8	B7; queue attribution for PCIR; RCST dispute flag for PCDR	

R2 - Actionable insight

ID	Build item	Scope	Capabilities	Depends on	Estimate
B9	Drill-down extraction	Per-leaf prompt configuration (what to extract for this node); transcript chunking so only the relevant section goes to the leaf prompt; results written to the drill-down structure. Order: eligibility/benefits, provider search, toolkit. Each reviewed with the business on sample output.	F5, F26	B1, B4, B5; per-node specs agreed with business	
B10	Targeted reprocessing	Re-run categorization/extraction for interactions under a changed node (or date range) rather than everything.	F10	B4, B9; job scheduler; LLM cost budget	
B11	Interactive reporting UI	Drill-down screens: category > subcategory > leaf attributes, filterable by caller type, group, program type, date.	F16	B7; UI framework decision	
B12	BI and trend measures	Curated Snowflake views for Power BI / Tableau; monthly counts, trends and call-level metrics derived in ETL (not stored operationally).	F13, F27, F30	B7; warehouse engineering	

R3 - Curate and discover

ID	Build item	Scope	Capabilities	Depends on	Estimate
B13	Business admin UI	Manage sets, versions and applicability; propose and approve taxonomy changes; audit trail.	F12	B6; authorization model	
B14	Category discovery	Periodic clustering over de-identified transcripts; candidate nodes surfaced into the editor for curation.	F11	B5, B13; embedding/clustering service; batch window	
B15	Chatbot sessions	Categorize chatbot transcripts with the same pipeline; add outcome tracking (fulfilled / not / escalated), mapped to tool calls where available.	F20, F29	Chatbot in production; session transcripts	

R4 - Explore

ID	Build item	Scope	Capabilities	Depends on	Estimate
B16	Chat with data	Agent over the Snowflake semantic layer that generates queries / analysis from natural-language questions; shared with the wider Snowflake initiative.	F15	Snowflake semantic layer; model with code/SQL generation	
B17	Ad-hoc mining	Keyword search and user-defined filters / time windows over transcripts and findings.	F14	Indexed de-identified transcripts	
B18	Routing and call-flow insight	Original queue vs actual group; cross-plan transfers; effect of IVR/routing changes.	F17, F18	Original-queue export from Genesys; definition agreed (O4)	
B19	Predictive and e-mail	Volume forecasting for conversions; categorization of e-mail inquiries.	F19, F20	R1-R3 history; e-mail source	

Design decision: how drill-down results are stored

Constraint: the schema must not change when a leaf's extraction list changes. Two options were discussed.

Criterion	Option A - JSON payload on the question record	Option B - key-value attribute table
Schema stability	None. One column holds whatever the leaf prompt produces.	None. Rows are added, not columns.
Querying in Oracle	JSON functions; adequate for operational lookups, awkward for aggregation.	Plain SQL; straightforward aggregation.
Snowflake path	Flatten in ETL (JSON is native to Snowflake; VARIANT or LATERAL FLATTEN).	Already row-shaped; direct load.
Validation	Needs a per-leaf schema and write-time validation, or payloads drift.	Names/types still need a spec; same discipline, enforced per row.
Reprocessing	Replace one payload per question.	Delete and reinsert N rows per question.
Row volume	One row per question.	Up to tens of rows per question; at 1,500+ calls/day this grows quickly.
Matches review direction	Yes ('maybe the transactional piece is just a big blob of JSON').	Discussed as the alternative.

Recommendation: **Option A**, with three safeguards.

- Each taxonomy leaf carries an **extraction spec** (attribute names, types, prompt instruction, spec version) stored as configuration, not DDL. The pipeline validates the JSON against the spec before writing and stores the spec version on the record.
- The analytics tables are **operational only**; aggregation and trending happen in Snowflake, where the JSON is flattened by ETL. This is the split the review agreed on.
- If a particular leaf becomes heavily queried operationally, its attributes can be promoted to a key-value side table later without touching the JSON contract.

Why not B: it does not buy schema stability beyond what A already gives, it multiplies row volume, and it still needs a spec to be meaningful. A keeps the transactional store simple and leaves structure to the warehouse.

Open items (updated)

ID	Item	Status
O1	Where drill-down attributes are stored	Recommendation above; confirm with reporting engineering.
O2	Interaction vs case anchoring	Resolved : interaction.
O3	Which reports each release supports	Resolved : see Release overview and B8 / B11 / B12.
O4	Definition of 'call-flow change effectiveness'	Open; R4 item, not blocking.
O5	Original-queue data from Genesys	Open; affects PCIR attribution (R1) and B18 (R4). Request export path early.
O6	Active flag vs latest version on sets	Open; decide during B1 design.
O7	Channel list (outbound, mail)	Open; not blocking.
O8	Separate sets for member vs provider	Resolved : separate subtrees under caller type.
O9	Group/subgroup on the interaction	Superseded : joined in Snowflake, not stored in analytics tables.
O10	New - De-identification approach	Evaluate Azure redaction service; security sign-off needed before de-identified data is used with AI tools.